

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Practicum 3: Fitting data and
09/19/2025

1. Practice fitting data with models (**35 points total**).

a. (**5 points**) Read in the data from file `practicum3_1.dat`. The file contains two sets of data. Model 1 data are made using function $f(x) = 3.2x + 1.2$ and adding some noise to the data, and Model 2 are made using function $f(x) = 3.2x^2 + 1.2$. Start by plotting your data of Model 1.

b. (**10 points**) Use the routine `linfit` (see your demos) to fit a linear model to the data. Assume that the data points each have an uncertainty in y of 0.5 (this is important, so explain why it makes a difference; how does your fit change if you have a σ_y of 0.2 or 0.9?) What are the fitted parameters of the model? Are they within 3σ of the parameters of the true line (the one we used to make the data)?

c. (**10 points**) What is the probability that you would get a higher χ^2 (i.e., a worse fit) by chance, if the data came from the fitted model and had the stated random error level? Plot the data and model, with reasonable axis labels and title. Save as an appropriately named PNG file.

d. (**10 points**) Repeat the steps b and c, but now for the second dataset (the quadratic data). Does the `linfit` model fit the data? Give two reasons why or why not. How can you fit the data?

2. This is the start of Problem 2 of your HW5. Work on this in class and get feedback on the fly, so that you can make the function of Problem 3 easier.

Sigma clipping, the technique of removing data points that are further than a few σ from your median, is a technique that allows you to clean data from outliers that shouldn't be there. For example, you could use sigma clipping to remove pixels that should contain only 'background' noise in your image but don't because your target star saturated its pixel and overflowed. Here, you will code your own sigma clipping to understand how the process works.

- a. (**10 points**) Create an array with 400 elements. The first 396 elements of the array should contain 396 draws from a Poisson distribution for 10,000 photons (i.e., the expected number of events occurring in a fixed-time interval should be 10,000). The last 4 elements should contain draws from the uniform distribution between 0 and 10^6 .

This sample represents data from a CCD with just 1% bad pixels. Print the mean and median of the sample and put the result in your file in comments. Which is closer to N ?

Tip: don't reinvent the wheel! There are numpy functions that calculate these two distributions when you give them as input the parameters we provide here.

- b. **(10 points)** Calculate the standard deviation, σ , for your sample.

Then, you will mask your data. Create a subsample including only points that are within 5σ of the median (you can do this in one line; check our intro to python lecture for masking of data).

Print the new sample's mean, median, and standard deviation. What has happened?

Continue to your HW5!